

Status Note: Vector-Valued Lipschitz Diameter Questions

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Status. `literature_already_answered` (scoped). This note records later literature answers to selected questions from Becerra Guerrero–Lopez-Perez–Rueda Zoca, arXiv:1512.03558, *Octahedrality in Lipschitz-free Banach spaces* [1].

Source questions. In Section 3 of [1], the authors ask whether, under the hypotheses of their Theorem 2.4, $\text{Lip}(M, X^*)$ satisfies the strong diameter two property. The same section asks for assumptions on a metric space M and a non-zero Banach space X ensuring that the vector-valued free space $\mathcal{F}(M, X) = \mathcal{F}(M) \widehat{\otimes}_\pi X$ has slice-D2P, D2P, or SD2P. In later notation these are cited as Questions 3.1 and 3.3 of [1].

Supporting answer. Langemets–Rueda Zoca, arXiv:1905.09061, *Octahedral norms in duals and biduals of Lipschitz-free spaces* [2], explicitly cites [1]. Theorem 2.4 of [2] proves that if the pair (M, X) has the contraction-extension property and M is unbounded or is not uniformly discrete, then $\text{Lip}(M, X)$ has the symmetric strong diameter two property. The paper states immediately after the theorem that this improves [1, Theorem 2.4] and gives a positive answer to [1, Question 3.1].

Theorem 3.1 of [2] proves that if $\text{Lip}(M)$ is octahedral, then $\text{Lip}(M, X^*)$ is octahedral for every Banach space X . Corollary 3.2 then states that, for a metric space M , the following are equivalent:

$\mathcal{F}(M, X)$ has the SD2P for every Banach space X ,

$\mathcal{F}(M)$ has the SD2P, $\mathcal{F}(M)$ has no strongly exposed point.

The supporting paper states that this corollary answers [1, Question 3.3].

Scope limitation. This packet is not a new proof. It records an already-known literature answer. The clean criterion above is for the SD2P component of the vector-valued free-space question. The source question also mentions slice-D2P and D2P; this packet does not claim a full classification of those weaker variants beyond consequences of the SD2P result.

References

- [1] J. Becerra Guerrero, G. Lopez-Perez, and A. Rueda Zoca, *Octahedrality in Lipschitz-free Banach spaces*, arXiv:1512.03558.
- [2] J. Langemets and A. Rueda Zoca, *Octahedral norms in duals and biduals of Lipschitz-free spaces*, arXiv:1905.09061.